

AI MACHINE HEALTH DASHBOARD - GUIDE SUMMARY

Project Page Created

- Page Name: AI Machine Health Dashboard
- Route: /ai-dashboard
- Main File: src/Pages/AIMachineHealth/AIMachineHealthDashboard.jsx
- Sidebar integrated: AI Dashboard entry added

What the AI Module Does

- Real-time machine health monitoring
- Failure risk prediction using:
temperature, pressure, vibration, spindle speed, current consumption,
wind speed, battery degradation
- Detects and flags:
overheating, tool damage, gearbox failure, full system issue,
crack detection (placeholder classifier)
- Future risk forecast:
next 1h, 24h, 7d risks with confidence
- Health status classification:
Healthy / Warning / Critical
- AI assistant insights + recommendation actions
- Equipment search with full related details

AI Approach Used

Hybrid pipeline:

1) API Inference Mode:

If VITE_AI_MACHINE_HEALTH_API_URL is connected, frontend calls backend for prediction/inference.

2) Local Fallback AI Mode:

Rule-based inference engine + synthetic telemetry provider, used when backend is unavailable.

This makes the module reliable both online and offline.

Core AI Logic

- Anomaly detection (threshold + trend behavior)
- Failure classification (status/severity mapping)
- Future failure projection with confidence
- Health score computation (0-100)
- Recommendation generation

Primary Output JSON Shape

- healthScore
- currentStatus
- anomalies[]
- alerts[]
- predictedFailures[]
- recommendations[]
- lastUpdated

Key Frontend Files

- src/Pages/AIMachineHealth/AIMachineHealthDashboard.jsx
- src/Pages/AIMachineHealth/useMachineHealthData.js

- src/Pages/AlMachineHealth/AlMachineHealthDashboard.css

AI Components

- src/Components/AlMachineHealth/HealthCard.jsx
- src/Components/AlMachineHealth/SensorChart.jsx
- src/Components/AlMachineHealth/SensorGraphModal.jsx
- src/Components/AlMachineHealth/HealthScoreGauge.jsx
- src/Components/AlMachineHealth/FailureForecast.jsx
- src/Components/AlMachineHealth/EventTimeline.jsx
- src/Components/AlMachineHealth/AIAssistantPanel.jsx
- src/Components/AlMachineHealth/AlertTable.jsx

Service / AI Layer Files

- src/Services/AlMachineHealth/machineHealthService.js
- src/Services/AlMachineHealth/ruleEngine.js
- src/Services/AlMachineHealth/schema.js
- src/Services/AlMachineHealth/mockProvider.js
- src/Services/AlMachineHealth/constants.js

Optional Python Backend Files

- ai/api/machine_health_api.py
- ai/model/machine_health_engine.py

Highlights Added in UI

- Animated/glow sensor lines with gradient fills
- Modal graph view per sensor (Open graph)
- KPI cards with trend sparklines and arrows
- AI alert intelligence cards (interactive diagnostics/recommendation)
- AI insight panel with Ask AI and Voice state
- Compact bottom widgets (CPU, latency, uptime, etc.)
- Equipment Condition Search with full detail drilldown

Validation Status

- Lint checks passed after implementation iterations
- Dark mode and light mode compatibility handled
- API + fallback behavior supported

Generated on: 2026-05-21 17:47:33